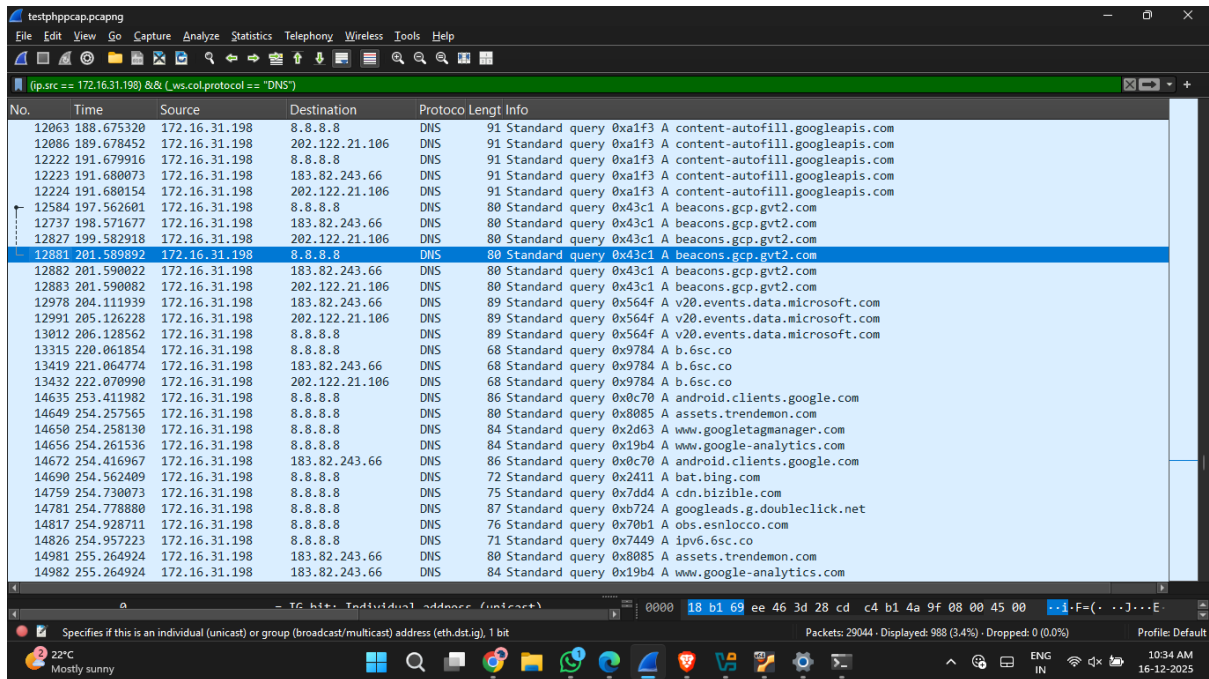


wireShark

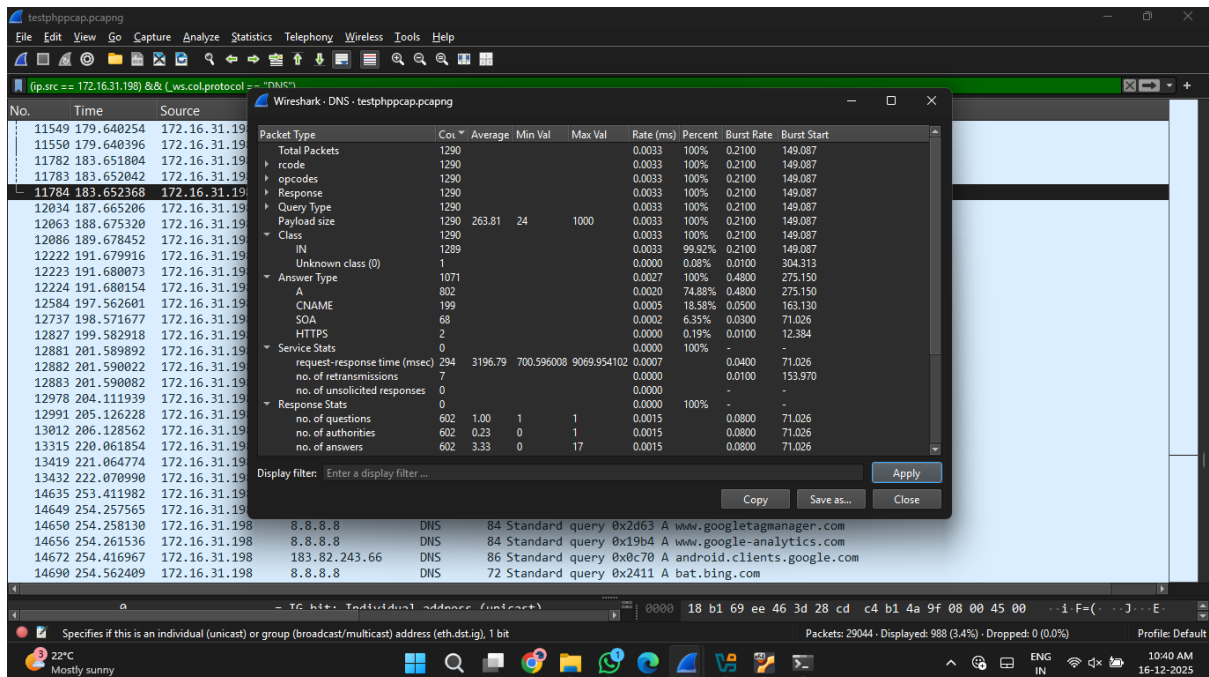
Task:1

1



No.	Time	Source	Destination	Protocol	Length	Info
12063	188.675320	172.16.31.198	8.8.8.8	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12086	189.678452	172.16.31.198	202.122.21.106	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12222	191.679916	172.16.31.198	8.8.8.8	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12223	191.680073	172.16.31.198	183.82.243.66	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12224	191.680154	172.16.31.198	202.122.21.106	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12584	197.562601	172.16.31.198	8.8.8.8	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12737	198.571677	172.16.31.198	183.82.243.66	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12827	199.582918	172.16.31.198	202.122.21.106	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12881	201.589892	172.16.31.198	8.8.8.8	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12882	201.590022	172.16.31.198	183.82.243.66	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12883	201.590082	172.16.31.198	202.122.21.106	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12978	204.111939	172.16.31.198	183.82.243.66	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
12991	205.126228	172.16.31.198	202.122.21.106	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
13012	206.128562	172.16.31.198	8.8.8.8	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
13315	220.061854	172.16.31.198	8.8.8.8	DNS	68	Standard query 0x9784 A b.6sc.co
13419	221.064774	172.16.31.198	183.82.243.66	DNS	68	Standard query 0x9784 A b.6sc.co
13432	222.070990	172.16.31.198	202.122.21.106	DNS	68	Standard query 0x9784 A b.6sc.co
14635	253.411982	172.16.31.198	8.8.8.8	DNS	86	Standard query 0x0c70 A android.clients.google.com
14649	254.257565	172.16.31.198	8.8.8.8	DNS	80	Standard query 0x8085 A assets.trendemon.com
14650	254.258130	172.16.31.198	8.8.8.8	DNS	84	Standard query 0x2d63 A www.googletagmanager.com
14656	254.261536	172.16.31.198	8.8.8.8	DNS	84	Standard query 0x19b4 A www.google-analytics.com
14672	254.416967	172.16.31.198	183.82.243.66	DNS	86	Standard query 0x0c70 A android.clients.google.com
14690	254.562409	172.16.31.198	8.8.8.8	DNS	72	Standard query 0x2411 A bat.bing.com
14759	254.730073	172.16.31.198	8.8.8.8	DNS	75	Standard query 0x7dd4 A cdn.bizible.com
14781	254.778800	172.16.31.198	8.8.8.8	DNS	87	Standard query 0xb724 A googleads.g.doubleclick.net
14817	254.928711	172.16.31.198	8.8.8.8	DNS	76	Standard query 0x70b1 A obs.esnlocco.com
14826	254.957223	172.16.31.198	8.8.8.8	DNS	71	Standard query 0x7449 A ipv6.6sc.co
14981	255.264924	172.16.31.198	183.82.243.66	DNS	80	Standard query 0x8085 A assets.trendemon.com
14982	255.264924	172.16.31.198	183.82.243.66	DNS	84	Standard query 0x19b4 A www.google-analytics.com

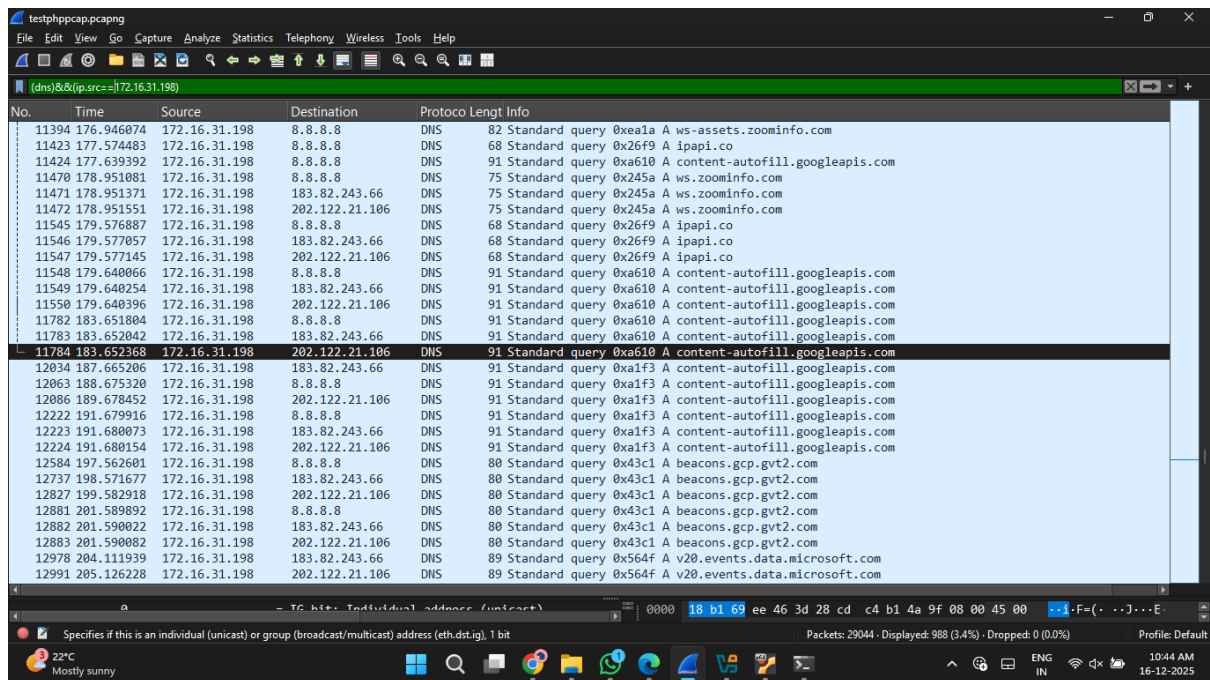
1.2



No.	Time	Source	Destination	Protocol	Length	Info
11549	179.640254	172.16.31.198	8.8.8.8	DNS	1290	Standard query 0x2d63 A www.googletagmanager.com
11550	179.640396	172.16.31.198	8.8.8.8	DNS	1290	Standard query 0x19b4 A www.google-analytics.com
11782	183.651804	172.16.31.198	183.82.243.66	DNS	1290	Standard query 0x0c70 A android.clients.google.com
11783	183.652042	172.16.31.198	8.8.8.8	DNS	1290	Standard query 0x2411 A bat.bing.com
11784	183.652368	172.16.31.198	8.8.8.8	DNS	1290	Standard query 0x2d63 A www.googletagmanager.com
12634	187.665206	172.16.31.198	8.8.8.8	DNS	1290	Standard query 0x2d63 A www.googletagmanager.com
12063	188.675320	172.16.31.198	8.8.8.8	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12086	189.678452	172.16.31.198	202.122.21.106	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12222	191.679916	172.16.31.198	8.8.8.8	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12223	191.680073	172.16.31.198	183.82.243.66	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12224	191.680154	172.16.31.198	202.122.21.106	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12584	197.562601	172.16.31.198	8.8.8.8	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12737	198.571677	172.16.31.198	183.82.243.66	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12827	199.582918	172.16.31.198	202.122.21.106	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12881	201.589892	172.16.31.198	8.8.8.8	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12882	201.590022	172.16.31.198	183.82.243.66	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12883	201.590082	172.16.31.198	202.122.21.106	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12978	204.111939	172.16.31.198	183.82.243.66	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
12991	205.126228	172.16.31.198	202.122.21.106	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
13012	206.128562	172.16.31.198	8.8.8.8	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
13315	220.061854	172.16.31.198	8.8.8.8	DNS	68	Standard query 0x9784 A b.6sc.co
13419	221.064774	172.16.31.198	183.82.243.66	DNS	68	Standard query 0x9784 A b.6sc.co
13432	222.070990	172.16.31.198	202.122.21.106	DNS	68	Standard query 0x9784 A b.6sc.co
14635	253.411982	172.16.31.198	8.8.8.8	DNS	86	Standard query 0x0c70 A android.clients.google.com
14649	254.257565	172.16.31.198	8.8.8.8	DNS	80	Standard query 0x8085 A assets.trendemon.com
14650	254.258130	172.16.31.198	8.8.8.8	DNS	84	Standard query 0x2d63 A www.googletagmanager.com
14656	254.261536	172.16.31.198	8.8.8.8	DNS	84	Standard query 0x19b4 A www.google-analytics.com
14672	254.416967	172.16.31.198	183.82.243.66	DNS	86	Standard query 0x0c70 A android.clients.google.com
14690	254.562409	172.16.31.198	8.8.8.8	DNS	72	Standard query 0x2411 A bat.bing.com

Packet Type	Cnt	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Total Packets	1290				0.0033	100%	0.2100	149.087
rcode	1290				0.0033	100%	0.2100	149.087
opcodes	1290				0.0033	100%	0.2100	149.087
Response	1290				0.0033	100%	0.2100	149.087
Query Type	1290				0.0033	100%	0.2100	149.087
Payload size	1290	263.81	24	1000	0.0033	100%	0.2100	149.087
Class	1290				0.0033	100%	0.2100	149.087
IN	1289				0.0033	99.92%	0.2100	149.087
Unknown class (0)	1				0.0000	0.08%	0.0100	304.313
Answer Type	1071				0.0027	100%	0.4800	275.150
A	802				0.0020	74.88%	0.4800	275.150
CNAME	199				0.0005	18.58%	0.0500	163.130
SOA	68				0.0002	6.35%	0.0300	71.026
HTTPS	2				0.0000	0.19%	0.0100	12.384
Service Stats	0				0.0000	100%	-	-
request-response time (msec)	294	3196.79	700.596008	9069.954102	0.0007	-	0.0400	71.026
no. of retransmissions	7				0.0000	-	0.0100	153.970
no. of unsolicited responses	0				0.0000	-	-	-
Response Stats	0				0.0000	100%	-	-
no. of questions	602	1.00	1	1	0.0015	-	0.0800	71.026
no. of authorities	602	0.23	0	1	0.0015	-	0.0800	71.026
no. of answers	602	3.33	0	17	0.0015	-	0.0800	71.026

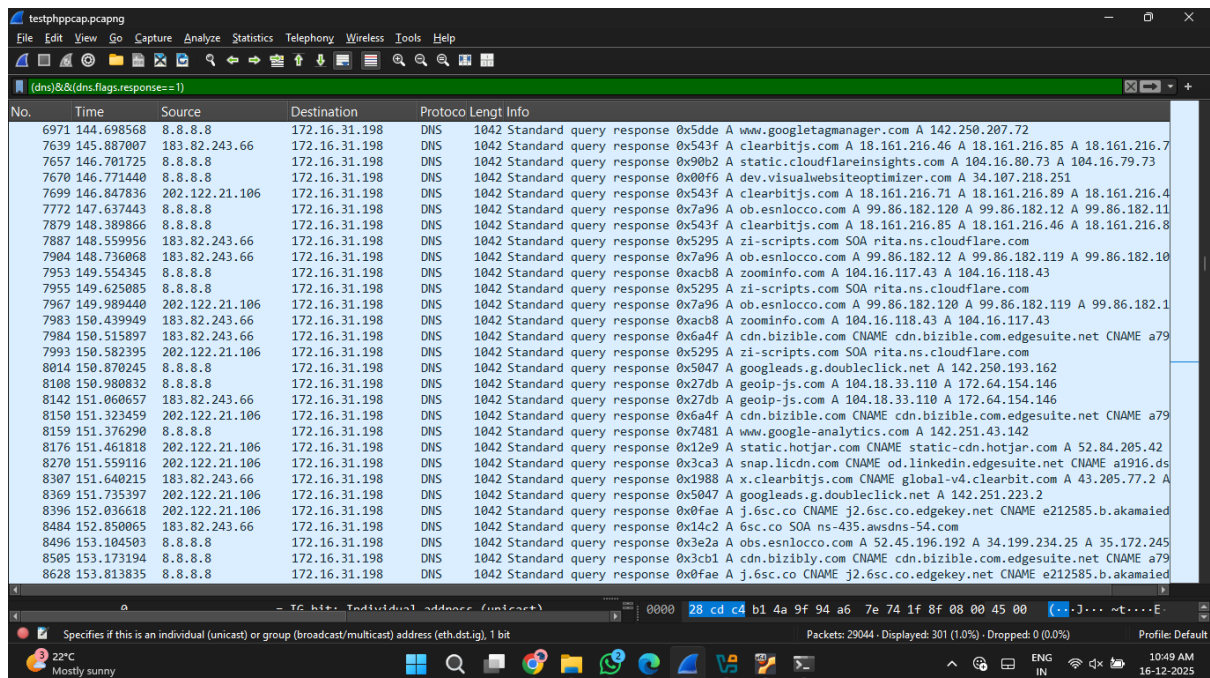
1.3



Wireshark capture of DNS traffic. The filter is `(dns)&&(ip.src==172.16.31.190)`. The packet list shows 29 packets, with the selected packet being a DNS query from 172.16.31.190 to 8.8.8.8.

No.	Time	Source	Destination	Protocol	Length	Info
11394	176.946074	172.16.31.190	8.8.8.8	DNS	82	Standard query 0xea1a A ws-assets.zoominfo.com
11423	177.574483	172.16.31.190	8.8.8.8	DNS	68	Standard query 0x26f9 A ipapi.co
11424	177.639392	172.16.31.190	8.8.8.8	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
11470	178.951081	172.16.31.190	8.8.8.8	DNS	75	Standard query 0x245a A ws.zoominfo.com
11471	178.951371	172.16.31.190	183.82.243.66	DNS	75	Standard query 0x245a A ws.zoominfo.com
11472	178.951551	172.16.31.190	202.122.21.106	DNS	75	Standard query 0x245a A ws.zoominfo.com
11545	179.576887	172.16.31.190	8.8.8.8	DNS	68	Standard query 0x26f9 A ipapi.co
11546	179.577057	172.16.31.190	183.82.243.66	DNS	68	Standard query 0x26f9 A ipapi.co
11547	179.577145	172.16.31.190	202.122.21.106	DNS	68	Standard query 0x26f9 A ipapi.co
11548	179.640066	172.16.31.190	8.8.8.8	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
11549	179.640254	172.16.31.190	183.82.243.66	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
11550	179.640396	172.16.31.190	202.122.21.106	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
11782	183.651804	172.16.31.190	8.8.8.8	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
11783	183.652042	172.16.31.190	183.82.243.66	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
11784	183.652368	172.16.31.190	202.122.21.106	DNS	91	Standard query 0xa610 A content-autofill.googleapis.com
12034	187.665206	172.16.31.190	183.82.243.66	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12063	188.675320	172.16.31.190	8.8.8.8	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12086	189.678452	172.16.31.190	202.122.21.106	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12222	191.679916	172.16.31.190	8.8.8.8	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12223	191.680073	172.16.31.190	183.82.243.66	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12224	191.680154	172.16.31.190	202.122.21.106	DNS	91	Standard query 0xa1f3 A content-autofill.googleapis.com
12584	197.562681	172.16.31.190	8.8.8.8	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12737	198.571677	172.16.31.190	183.82.243.66	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12827	199.582918	172.16.31.190	202.122.21.106	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12881	201.589892	172.16.31.190	8.8.8.8	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12882	201.590022	172.16.31.190	183.82.243.66	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12883	201.590082	172.16.31.190	202.122.21.106	DNS	80	Standard query 0x43c1 A beacons.gcp.gvt2.com
12978	204.111939	172.16.31.190	183.82.243.66	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com
12991	205.126228	172.16.31.190	202.122.21.106	DNS	89	Standard query 0x564f A v20.events.data.microsoft.com

1.4



Wireshark capture of DNS traffic. The filter is `(dns)&&(dns.flags.response=1)`. The packet list shows 301 packets, with the selected packet being a DNS response from 8.8.8.8 to 172.16.31.190.

No.	Time	Source	Destination	Protocol	Length	Info
6971	144.698568	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x5dde A www.googletagmanager.com A 142.250.207.72
7639	145.887007	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x543f A clearbitjs.com A 18.161.216.46 A 18.161.216.85 A 18.161.216.7
7657	146.701725	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x90b2 A static.cloudflareinsights.com A 104.16.80.73 A 104.16.79.73
7670	146.771440	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x00f6 A dev.visualwebsiteoptimizer.com A 34.107.218.251
7699	146.847836	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x543f A clearbitjs.com A 18.161.216.71 A 18.161.216.89 A 18.161.216.4
7772	147.637443	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x7a96 A ob.esnlocco.com A 99.86.182.120 A 99.86.182.12 A 99.86.182.11
7879	148.389866	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x543f A clearbitjs.com A 18.161.216.85 A 18.161.216.46 A 18.161.216.8
7887	148.559956	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x5295 A zi-scripts.com SOA rita.ns.cloudflare.com
7904	148.736068	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x7a96 A ob.esnlocco.com A 99.86.182.12 A 99.86.182.119 A 99.86.182.10
7953	149.554345	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0xacb8 A zoominfo.com A 104.16.117.43 A 104.16.118.43
7955	149.625085	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x5295 A zi-scripts.com SOA rita.ns.cloudflare.com
7967	149.989440	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x7a96 A ob.esnlocco.com A 99.86.182.120 A 99.86.182.119 A 99.86.182.1
7983	150.439949	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0xacb8 A zoominfo.com A 104.16.118.43 A 104.16.117.43
7984	150.515897	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x6a4f A cdn.bizible.com CNAME cdn.bizible.com.edgesuite.net CNAME a79
7993	150.582395	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x5295 A zi-scripts.com SOA rita.ns.cloudflare.com
8014	150.870245	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x5047 A googleads.g.doubleclick.net A 142.250.193.162
8108	150.980832	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x27db A geoip-js.com A 104.18.33.110 A 172.64.154.146
8142	151.060657	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x27db A geoip-js.com A 104.18.33.110 A 172.64.154.146
8150	151.323459	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x6a4f A cdn.bizible.com CNAME cdn.bizible.com.edgesuite.net CNAME a79
8159	151.376290	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x7481 A www.google-analytics.com A 142.251.43.142
8176	151.461818	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x12e9 A static.hotjar.com CNAME static-cdn.hotjar.com A 52.84.205.42
8270	151.559116	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x3ca3 A snap.l1cdn.com CNAME od.linkedin.edgesuite.net CNAME a1916.ds
8307	151.640215	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x1988 A x.clearbitjs.com CNAME global-v4.clearbit.com A 43.205.77.2 A
8369	151.735397	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x5047 A googleads.g.doubleclick.net A 142.251.223.2
8396	152.036618	202.122.21.106	172.16.31.190	DNS	1042	Standard query response 0x0fae A j.6sc.co SOA ns-435.awsdns-54.com
8484	152.050065	183.82.243.66	172.16.31.190	DNS	1042	Standard query response 0x14c2 A 6sc.co SOA ns-435.awsdns-54.com
8496	153.104593	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x3a2a A ob.esnlocco.com A 52.45.196.192 A 34.199.234.25 A 35.172.245
8505	153.173194	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x3cb1 A cdn.bizible.com CNAME cdn.bizible.com.edgesuite.net CNAME a79
8628	153.813835	8.8.8.8	172.16.31.190	DNS	1042	Standard query response 0x0fae A j.6sc.co CNAME j2.6sc.co.edgesuite.net CNAME e212585.b.akamaied

```
testphpccap.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

[Icons]

http

No. Time Source Destination Protocol Length Info
1320 29.309211 172.16.31.198 44.228.249.3 HTTP 536 GET /login.php%20use%20this%20to%20generate%20traffic HTTP/1.1
1324 29.545988 44.228.249.3 172.16.31.198 HTTP 428 HTTP/1.1 404 Not Found (text/html)

Frame 3701: Packet, 703 bytes on wire (5624 bits), 703 bytes captured (5624 bits) on interface 0
Ethernet II, Src: ChongqingFugBi-4a:9f (28:cdc4:b1:4a:9f), Dst: SonicWall-ee:46:3d (08:00:27:2c:46:3d)
Internet Protocol Version 4, Src: 172.16.31.198, Dst: 44.228.249.3
Transmission Control Protocol, Src Port: 56634, Dst Port: 80, Seq: 1862, Ack: 15080, Len: 100
Hypertext Transfer Protocol
POST /userinfo.php HTTP/1.1\r\n
Host: testphp.vulnweb.com\r\n
Connection: keep-alive\r\n
Content-Length: 24\r\n
Cache-Control: max-age=0\r\n
Origin: http://testphp.vulnweb.com\r\n
Content-Type: application/x-www-form-urlencoded\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/104.0.5112.101 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+xml,*/*;q=0.8\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
[Response in frame: 3737]
[Full request URI: http://testphp.vulnweb.com/userinfo.php]
Content Type: 24 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded

Bytes 265-294: Request line (http.request_line)

Packets: 29044 - Displayed: 72 (0.2%) - Dropped: 0 (0.0%)
Profile: Default
22°C Mostly sunny 10:56 AM 16-12-2025
```

The screenshot shows a Windows desktop with a Wireshark packet capture window open. The main window is titled "Wireshark - HTTP / Packet Counter - testphp pcap.pcapng". The packet list pane shows a list of captured packets, with the first packet being an HTTP GET request. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane shows the raw data of the selected packet. The status bar at the bottom indicates the capture is on the "HTTP Host (http.host), 27 bytes".

Packet List:

No.	Time	Source
1320	29.389211	172.16.31.19
1324	29.545988	44.228.249.3
1334	29.854479	172.16.31.19
1346	30.089754	44.228.249.3
1686	37.240909	172.16.31.19
1717	37.519181	172.16.31.19
2042	42.428832	172.16.31.19
2049	42.897224	44.228.249.3
2053	42.895791	172.16.31.19
2055	42.899773	172.16.31.19
2102	43.216002	44.228.249.3
2111	43.224833	44.228.249.3
2322	49.178894	172.16.31.19
2451	53.586655	172.16.31.19
2454	53.835372	44.228.249.3
2613	55.210872	172.16.31.19
2648	55.459673	44.228.249.3
2805	60.718316	172.16.31.19

Packet Details:

- Ethernet II, Src: ChongqingFugu (08:00:0c:21:a8:55), Dst: 172.16.31.19
- Internet Protocol Version 4, Src: 172.16.31.19, Dst: 44.228.249.3
- Transmission Control Protocol, Src Port: 54321, Dst Port: 80
- Hypertext Transfer Protocol
 - POST /userinfo.php HTTP/1.1
 - Host: testphp.vulnweb.com
 - Connection: keep-alive
 - Content-Length: 24
 - Cache-Control: max-age=0
 - Origin: http://testphp.vulnweb.com
 - Content-Type: application/x-www-form-urlencoded

Packet Bytes:

```

0000  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0010  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0020  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100  45 2f 00 00 00 00 00 00 00 00 00 00 00 00 00 00
  
```

4

testphppcap.pcapng

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(ip.src==172.16.31.198)&&tcp.flags.syn==1&&tcp.flags.ack==0

No.	Time	Source	Destination	Protocol	Length	Info
459	13.223751	172.16.31.198	106.51.45.83	TCP	66	61008 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
594	17.350484	172.16.31.198	216.239.38.120	TCP	66	52777 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
595	17.351011	172.16.31.198	216.239.38.120	TCP	66	50348 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
596	17.351369	172.16.31.198	216.239.38.120	TCP	66	54496 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
599	17.356514	172.16.31.198	74.125.24.84	TCP	66	54155 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
608	17.382803	172.16.31.198	216.239.38.120	TCP	66	57113 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
609	17.383384	172.16.31.198	216.239.38.120	TCP	66	49841 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
610	17.383917	172.16.31.198	216.239.38.120	TCP	66	62725 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
611	17.384441	172.16.31.198	216.239.38.120	TCP	66	54118 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1303	29.071481	172.16.31.198	44.228.249.3	TCP	66	59158 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1304	29.072372	172.16.31.198	44.228.249.3	TCP	66	55848 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1317	29.308622	172.16.31.198	44.228.249.3	TCP	66	59774 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1476	32.330800	172.16.31.198	142.251.43.138	TCP	66	58259 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1679	36.807194	172.16.31.198	92.223.116.252	TCP	66	58260 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1691	37.360794	172.16.31.198	92.223.116.252	TCP	66	58261 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1999	42.188644	172.16.31.198	44.228.249.3	TCP	66	56634 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2000	42.189220	172.16.31.198	44.228.249.3	TCP	66	54478 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2052	42.883058	172.16.31.198	142.251.220.99	TCP	66	63208 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Frame 459: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

Ethernet II, Src: ChongqingFug_b1:4a:9f (28:cd:c4:b1:4a:9f), Dst: Sonicwall_e:46:3d (08:00:27:00:00:00)

Internet Protocol Version 4, Src: 172.16.31.198, Dst: 106.51.45.83

Transmission Control Protocol, Src Port: 61008, Dst Port: 443, Seq: 0, Len: 0

testphppcap.pcapng

Packets: 29044 · Displayed: 174 (0.6%) · Dropped: 0 (0.0%)

Profile: Default

24°C Mostly cloudy

11:41 AM 16-12-2025

5

testphppcap.pcapng

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Wireshark · Packet 634 · testphppcap.pcapng

tcp.flags.syn==1&&

No.	Time	Source	Destination	Protocol	Length	Info
460	13.228	172.16.31.198	106.51.45.83	TCP	66	61008 → 443 [ACK] Seq=0 Win=65535 Len=0
600	17.379	172.16.31.198	216.239.38.120	TCP	66	50348 → 443 [ACK] Seq=0 Win=65535 Len=0
601	17.379	172.16.31.198	216.239.38.120	TCP	66	54496 → 443 [ACK] Seq=0 Win=65535 Len=0
612	17.384	172.16.31.198	74.125.24.84	TCP	66	54155 → 443 [ACK] Seq=0 Win=65535 Len=0
626	17.422	172.16.31.198	216.239.38.120	TCP	66	57113 → 443 [ACK] Seq=0 Win=65535 Len=0
628	17.422	172.16.31.198	216.239.38.120	TCP	66	49841 → 443 [ACK] Seq=0 Win=65535 Len=0
634	17.430	172.16.31.198	106.51.45.83	TCP	66	61008 → 443 [ACK] Seq=0 Win=65535 Len=0
640	17.445	172.16.31.198	216.239.38.120	TCP	66	50348 → 443 [ACK] Seq=0 Win=65535 Len=0
646	17.452	172.16.31.198	216.239.38.120	TCP	66	54496 → 443 [ACK] Seq=0 Win=65535 Len=0
1315	29.307	172.16.31.198	44.228.249.3	TCP	66	59158 → 80 [ACK] Seq=0 Win=65535 Len=0

Frame 634: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

Ethernet II, Src: ChongqingFug_b1:4a:9f (28:cd:c4:b1:4a:9f), Dst: Sonicwall_e:46:3d (08:00:27:00:00:00)

Internet Protocol Version 4, Src: 172.16.31.198, Dst: 106.51.45.83

Transmission Control Protocol, Src Port: 61008, Dst Port: 443, Seq: 0, Len: 0

Window: 65535

[Calculated window size: 65535]

Checksum: 0x5440 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window

[Timestamps]

[Time since first frame in this TCP stream: 46.853000 milliseconds]

[Time since previous frame in this TCP stream: 46.853000 milliseconds]

[SEQ/ACK analysis]

[This is an ACK to the segment in frame: 610]

[The RTT to ACK the segment was: 46.853000 milliseconds]

[RTT: 46.966000 milliseconds]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

0000 28 cd c4 b1 4a 9f 18 b1 69 ee 46 3d 08 00 45 00 (...J...i.F...E.

Encapsulation type (frame encaps. type)

Show packet bytes Layout: Vertical (Stacked)

Flags: 0x00

Window: 65535

[Calculated window size: 65535]

Checksum: 0x5440 [unverified]

Source Port (tcp.srcport), 2 bytes

testphppcap.pcapng

Packets: 29044 · Displayed: 175 (0.6%) · Dropped: 0 (0.0%)

Profile: Default

24°C Mostly cloudy

11:50 AM 16-12-2025

The screenshot shows the Wireshark network protocol analyzer interface. The main pane displays the details of a selected packet (Frame 2102), which is an HTTP GET request for the file `/style.css`. The packet is captured on the `\Device\NPF_{27BFA7C5-E844-4A68-...}` interface. The details pane shows the following structure:

- Ethernet II**: Src: Sonicwall_ee:46:3d (18:b1:69:ee:46:3d), Dst: ChongqingFug_b1:4a:9f (28:cd:c4:b1:4a:9f)
- Internet Protocol Version 4**: Src: 44.228.249.3, Dst: 172.16.31.198
- Transmission Control Protocol**: Src Port: 80, Dst Port: 54478, Seq: 7179, Ack: 769, Len: 1102
- [5 Reassembled TCP Segments (5721 bytes): #2092(239), #2093(1460), #2095(1460), #2096(1460), #2102(1102)]**
- Hypertext Transfer Protocol**:
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Server: nginx/1.19.0
 - Date: Tue, 16 Dec 2025 04:07:07 GMT
 - Content-Type: text/css
 - Content-Length: 5482
 - Last-Modified: Wed, 11 May 2011 10:27:48 GMT
 - Connection: keep-alive
 - ETag: "4dca64a4-156a"
 - Accept-Ranges: none
 - [Request in frame: 2053]
 - [Time since request: 320.211000 milliseconds]
 - [Request URI: /style.css]

The packet list on the left shows the selected packet (Frame 2102) and its corresponding Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol details. The packet bytes pane at the bottom shows the raw data of the packet, which is a reassembled TCP segment.

The image displays the Wireshark network traffic analysis interface. The main window is titled "testphpccap.pcapng". The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The top toolbar contains various icons for file operations, capture, and analysis.

The left pane shows the packet list with columns for No., Time, and Source. The selected packet is 2102, which is an HTTP 200 OK response from 44.228.249.3. The packet details pane on the right shows the structure of the packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane at the bottom shows the raw data of the packet, including the HTTP response status and headers.

The main pane displays the "Wireshark - Export - HTTP object list" window, which lists the contents of the HTTP response. The list includes various files and their sizes, such as login.php, favicon.ico, style.css, logo.gif, categories.php, artists.php, login.php, userinfo.php, signup.php, index.php, and categories.php. The list is filtered by "Content Type: All Content-Types".

The bottom status bar shows the current packet count (29044) and the dropped packet count (0). The system tray at the bottom right displays the date and time (12:10 PM, 16-12-2025) and the network status (ENG IN).

Task2:

1

The screenshot shows a Wireshark packet capture of testphpcap111.pcapng. The filter bar at the top is set to http.request.method==GET. The packet list pane displays 18 HTTP GET requests from 172.16.31.198 to 44.228.249.3. The packet details pane for packet 397 (No. 647) is selected, showing Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol details. The packet bytes pane shows the raw data of the packet. The status bar at the bottom indicates 2590 packets displayed, with 18 (0.7%) shown and 0 dropped.

No.	Time	Source	Destination	Protocol	Length	Info
559	9.374659	172.16.31.198	44.228.249.3	HTTP	497	GET /login.php HTTP/1.1
647	9.652676	172.16.31.198	44.228.249.3	HTTP	397	GET /style.css HTTP/1.1
649	9.663862	172.16.31.198	44.228.249.3	HTTP	449	GET /images/logo.gif HTTP/1.1
728	10.250683	172.16.31.198	44.228.249.3	HTTP	445	GET /favicon.ico HTTP/1.1
1639	37.616585	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/1.jpg&size=160 HTTP/1.1
1643	37.628764	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/2.jpg&size=160 HTTP/1.1
1671	37.868154	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/3.jpg&size=160 HTTP/1.1
1675	37.868405	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/4.jpg&size=160 HTTP/1.1
1684	37.864523	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/5.jpg&size=160 HTTP/1.1
1685	37.864898	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/6.jpg&size=160 HTTP/1.1
1688	37.865573	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/7.jpg&size=160 HTTP/1.1
1756	38.121947	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/1.jpg&size=160 HTTP/1.1
1757	38.122430	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/3.jpg&size=160 HTTP/1.1
1777	38.365618	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/2.jpg&size=160 HTTP/1.1
1788	38.370363	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/4.jpg&size=160 HTTP/1.1
1796	38.372950	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/5.jpg&size=160 HTTP/1.1
1797	38.373344	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/6.jpg&size=160 HTTP/1.1
1800	38.374047	172.16.31.198	44.228.249.3	HTTP	490	GET /showImage.php?file=../pictures/7.jpg&size=160 HTTP/1.1

2

The screenshot shows a Wireshark packet capture of testphpcap111.pcapng. The filter bar at the top is set to http.response.code==200. The packet list pane displays 20 HTTP 200 OK responses from 44.228.249.3 to 172.16.31.198. The packet details pane for packet 1342 (No. 645) is selected, showing Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol details. The packet bytes pane shows the raw data of the packet. The status bar at the bottom indicates 2590 packets displayed, with 36 (1.4%) shown and 0 dropped.

No.	Time	Source	Destination	Protocol	Length	Info
645	9.612569	44.228.249.3	172.16.31.198	HTTP	1342	HTTP/1.1 200 OK (text/html)
661	9.891381	44.228.249.3	172.16.31.198	HTTP	1156	HTTP/1.1 200 OK (text/css)
669	9.902817	44.228.249.3	172.16.31.198	HTTP	874	HTTP/1.1 200 OK (GIF89a)
741	10.490893	44.228.249.3	172.16.31.198	HTTP	948	HTTP/1.1 200 OK (image/x-icon)
1137	18.211398	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1142	18.374266	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1242	24.262030	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1253	24.446215	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1270	24.703994	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1304	24.838605	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1317	25.009736	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1337	25.204236	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1345	25.323102	44.228.249.3	172.16.31.198	HTTP	1071	HTTP/1.1 200 OK (text/html)
1373	28.252064	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1386	28.420598	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1400	28.656574	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1411	28.800561	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1425	28.951111	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1437	29.124679	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1446	29.249846	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1451	29.367866	44.228.249.3	172.16.31.198	HTTP	1070	HTTP/1.1 200 OK (text/html)
1561	35.848323	44.228.249.3	172.16.31.198	HTTP	1073	HTTP/1.1 200 OK (text/html)
1618	37.420992	44.228.249.3	172.16.31.198	HTTP	1419	HTTP/1.1 200 OK (text/html)
1636	37.505064	44.228.249.3	172.16.31.198	HTTP	1419	HTTP/1.1 200 OK (text/html)

3

Wireshark capture of an HTTP response. The packet list shows a single packet (No. 12442) at time 15.638712, source 23.62.12.40, destination 172.16.30.190, protocol HTTP, length 241 bytes. The packet details pane shows the following structure:

- Frame 12442: Packet, 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface \Device\NPF...
- Ethernet II, Src: Sonicwall_ee:46:3d (18:b1:69:ee:46:3d), Dst: AzureWaveTec_ad:a0:78 (1c:ce:51:ad:a0:78)
- Internet Protocol Version 4, Src: 23.62.12.40, Dst: 172.16.30.190
- Transmission Control Protocol, Src Port: 80, Dst Port: 54621, Seq: 1, Ack: 148, Len: 187
- Hypertext Transfer Protocol
 - Line-based text data: text/plain (1 lines)

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII portion shows the response status line: "200 OK Content-Length: 22 Date: Tue, 16 Dec 202 5:09:33 GMT".

4

Wireshark capture of an HTTP response. The packet list shows three packets (Nos. 661, 669, 741) at time 9.981381, source 44.228.249.3, destination 172.16.31.198, protocol HTTP, lengths 1156, 874, and 948 bytes respectively. The packet details pane shows the following structure:

- Frame 661: Packet, 1156 bytes on wire (9248 bits), 1156 bytes captured (9248 bits) on interface \Device\NPF...
- Ethernet II, Src: Sonicwall_ee:46:3d (18:b1:69:ee:46:3d), Dst: ChongqingFug_b1:4a:9f (08:00:27:00:00:00)
- Internet Protocol Version 4, Src: 44.228.249.3, Dst: 172.16.31.198
- Transmission Control Protocol, Src Port: 80, Dst Port: 49458, Seq: 7368, Ack: 787, Len: 1156
- Hypertext Transfer Protocol
 - Line-based text data: text/css (1 lines)

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII portion shows the response status line: "200 OK Content-Length: 1156 Date: Tue, 16 Dec 202 5:09:33 GMT".

Task3: 1 and 2

The screenshot shows a Wireshark capture of an HTTP GET request. The packet list on the left shows packet 2 as the selected packet. The packet details pane on the right shows the structure of the packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.600079	139.162.123.134	10.9.0.2	HTTP2	164	HTTP/1.1 101 Switching Protocols , SETTINGS[0]
3	0.600465	10.9.0.2	139.162.123.134	HTTP2	90	Magic
4	0.600541	10.9.0.2	139.162.123.134	HTTP2	93	SETTINGS[0]
5	0.600575	10.9.0.2	139.162.123.134	HTTP2	75	SETTINGS[0]
6	0.600596	139.162.123.134	10.9.0.2	HTTP2	342	HEADERS[1]: 200 OK, DATA[1] (text/plain)
7	0.600603	10.9.0.2	139.162.123.134	HTTP2	79	WINDOW_UPDATE[0]
8	0.601307	10.9.0.2	139.162.123.134	HTTP2	115	HEADERS[3]: GET /humans.txt
9	0.912304	139.162.123.134	10.9.0.2	HTTP2	75	SETTINGS[0]
10	0.916413	139.162.123.134	10.9.0.2	HTTP2	156	HEADERS[3]: 404 Not Found, DATA[3] (text/plain)

Frame 2: Packet, 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface 0
Ethernet II, Src: 8a:7d:40:9e:52:1b (8a:7d:40:9e:52:1b), Dst: 92:76:39:be:c1:81 (92:76:39:be:c1:81)
Internet Protocol Version 4, Src: 139.162.123.134, Dst: 10.9.0.2
Transmission Control Protocol, Src Port: 80, Dst Port: 58038, Seq: 1, Ack: 179, Len: 9
Hypertext Transfer Protocol
Hypertext Transfer Protocol 2

HyperText Transfer Protocol 2: Protocol

3

The screenshot shows a Wireshark capture of an HTTP 101 response. The packet list on the left shows packet 2 as the selected packet. The packet details pane on the right shows the structure of the packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.600079	139.162.123.134	10.9.0.2	HTTP2	164	HTTP/1.1 101 Switching Protocols , SETTINGS[0]
3	0.600465	10.9.0.2	139.162.123.134	HTTP2	90	Magic

Frame 2: Packet, 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface 0
Ethernet II, Src: 8a:7d:40:9e:52:1b (8a:7d:40:9e:52:1b), Dst: 92:76:39:be:c1:81 (92:76:39:be:c1:81)
Internet Protocol Version 4, Src: 139.162.123.134, Dst: 10.9.0.2
Transmission Control Protocol, Src Port: 80, Dst Port: 58038, Seq: 1, Ack: 179, Len: 9
Hypertext Transfer Protocol
Hypertext Transfer Protocol 2

HTTP/1.1 101 Switching Protocols\r\n\r\n

Response Version: HTTP/1.1
Status Code: 101
[Status Code Description: Switching Protocols]
Response Phrase: Switching Protocols
Connection: Upgrade\r\n\r\n

[Request in frame: 1]
[Time since request: 600.079000 milliseconds]
[Request URI: /robots.txt]
[Full request URI: http://mghttp2.org/robots.txt]

HyperText Transfer Protocol 2
Stream: SETTINGS, Stream ID: 0, Length 18
Length: 18
Type: SETTINGS (4)
Flags: 0x00
0... .. = Reserved: 0x0
.000 0000 0000 0000 0000 0000 0000 = Stream Identifier: 0
Settings - Max concurrent streams: 100
Settings - Initial Windows size: 1048576
Settings - Header table size: 8192

HTTP Response HTTP-Version (http.response.version), 8 bytes

